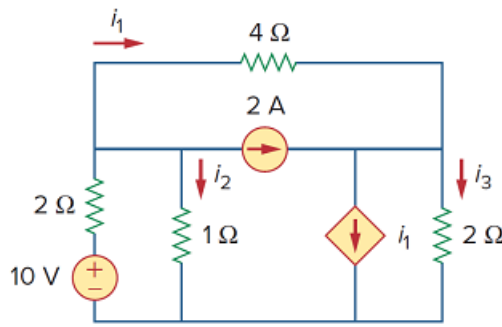


Problem

Use *PSpice* to determine currents i_1 , i_2 , and i_3 in the circuit of Fig. 3.36.

Figure 3.36:

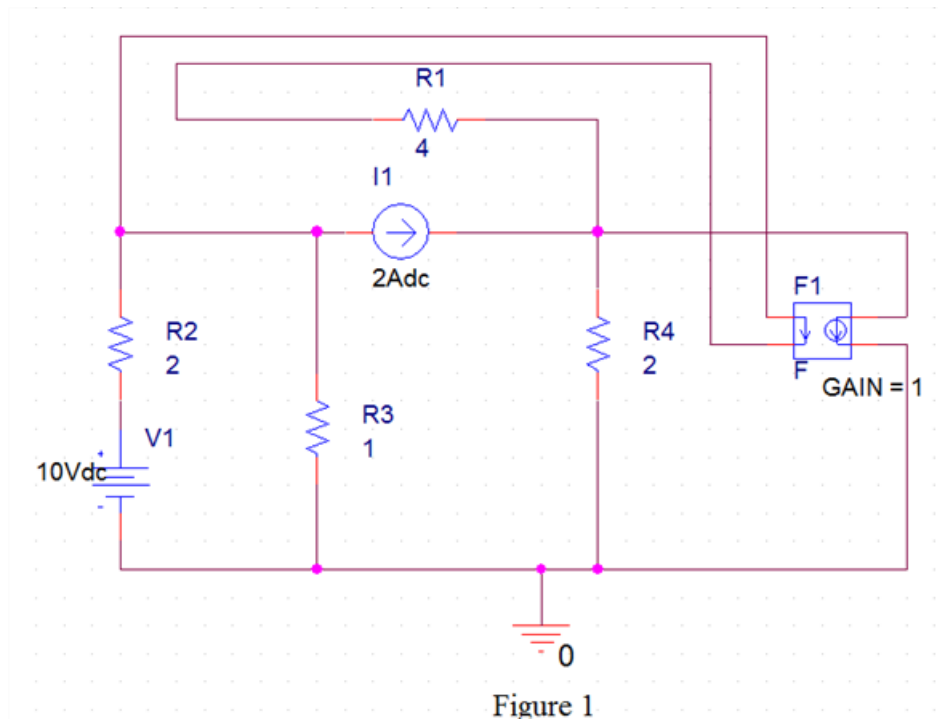


Step-by-step solution

Step 1 of 3

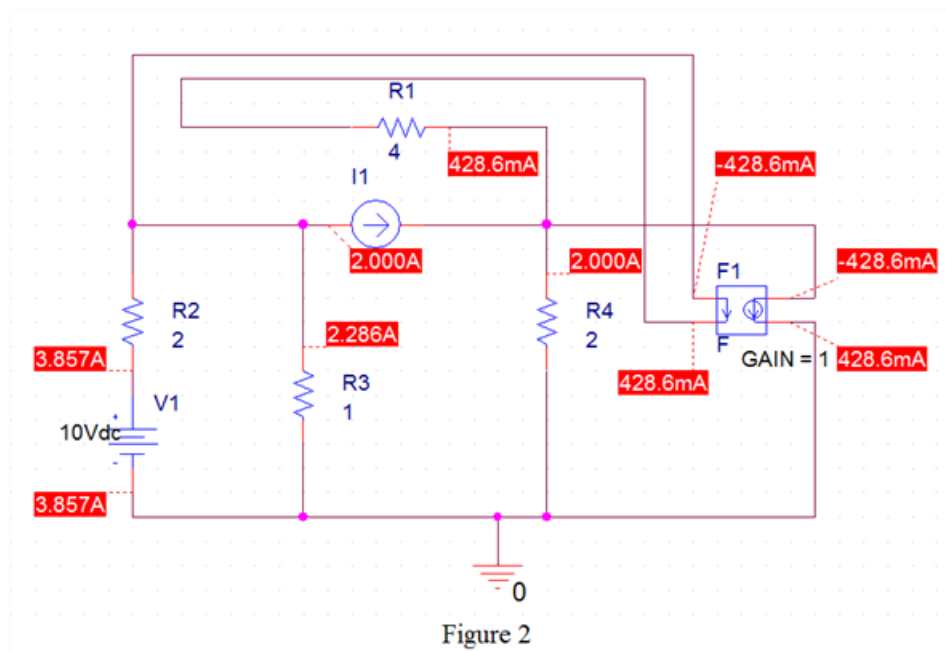
Refer to Figure 3.36 for the circuit diagram in the textbook.

Connect the circuit in the Pspice as shown in Figure 1. Use **P** key to select the components from the Part menu. Select **IDC** for the dc current source, **R** for the resistor, **VDC** for the dc voltage source, **F** for the current controlled current source, and **G** for the ground. Also double click on each component and assign the specified values to the components. Ensure that the gain of the current controlled current source is unity for the circuit.



Step 2 of 3

Save the settings and run the simulation using **F11** key. After completing the simulation, check the **Enable Bias Current Display**. Obtain the values of current flowing in each and every branch of the circuit as shown in Figure 2.



Step 3 of 3

Therefore, the values of the branch currents i_1 , i_2 , and i_3 are -0.4286 A , 2.286 A , and 2 A .